Faculty of Computer Applications and Information Technology
M. Sc. (IT) – Sem 3
Practicals using MERN
React Advance Tasks

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1. Design a Simple Signup Form with fields: Full Name, Email, Mobile Number.
   - Validate that none of the fields are empty.
   - Mobile Number must contain exactly 10 digits.
   - On submission, show the entered data below the form in JSON format.

2. Create an Employee Feedback Form with fields: Employee Name, Department, and Feedback.
   - All fields must be filled before submission.
   - Show an alert “Feedback Submitted” when the form is successfully submitted.
   - Clear all fields after submission.

3. Build a React Router-based app for a College Portal:
   / → Home Page (college name and motto)
   /courses → List of available courses
   /contact → Contact Form (Name + Email + Message)
   Add a navigation bar and a “Page Not Found” route for invalid URLs.

4. Develop a React Router app for a Restaurant Website:
   / → Home (Welcome note)
   /menu → List of dishes offered
   /book → Table booking form (Name + Date + Time)
   Include navigation links and a fallback 404 route.

5. Create a simple Todo List using Redux Toolkit to manage tasks.
   - Create a slice todoSlice with an initial state { todos: [] }.
   - Define reducers: addTodo and removeTodo.
   - Provide an input field to add a new todo.
   - Display all todos in a list with a delete button beside each.
   - Update the Redux store whenever a task is added or removed.

6. Build a simple theme toggle system using Redux Toolkit to switch between dark and light mode.
   - Create a slice themeSlice with an initial state { darkMode: false }.

- Define a reducer toggleTheme to change the mode.
- Add a button to toggle between dark and light modes.
- Display the current mode on the screen.
- Change background color based on the mode. (Optional)

7. Build a Temperature Controller App using Redux Toolkit.
   • Create a slice tempSlice with state { temperature: 20 }.
   • Define reducers: increaseTemp and decreaseTemp.
   • Add buttons to increase or decrease temperature.
   • Display the current temperature on screen.

8. Create a Feedback Collector App using Redux Toolkit.
   • Create a slice feedbackSlice with state { feedbacks: [] }.
   • Define reducer addFeedback.
   • Add input fields for Name and Feedback.
   • Display all submitted feedbacks below the form.

9. Create a slice pointsSlice with state { totalPoints: 0 }.
   • Define a reducer addPoints to increase points by a user-entered value.
   • Add an input field to enter the number of points to add.
   • Add a button labeled “Add Points”.
   • Display the updated total points on the screen.